Wireless Communication 02

Recall:

Computing devices communicate with each other to share information like files, images, videos and so on. We commonly use wired computing devices such as a mouse or a keyboard.

The 21st century has seen a rise in wireless connection systems. Many such technologies have been developed that allow users to interact with devices without the hassle of cables.

Data can be now transferred wirelessly via the internet.

In this lesson, we will learn how wireless devices connect to each other to share information.

2. Wireless Communication

Data Transfer Speed

Data transfer speed is the amount of data transferred per second. The most used metric is Mbps – Megabits per second. We know that 1 byte is 8 bits. So, a speed of 8 Mbps = 1 MB of data transfer per second.

1. Bluetooth

a. Bluetooth is a wireless technology that is used to transfer data between devices close to each other.

b. It is usually used in mobile phones and other portable devices like laptops.

- Range is the maximum physical distance between two networking devices so that the connection between them stays intact.
- The range of Bluetooth connectivity is between 0 to 50 metres. This is ideal for personal devices to share data with each other.

For example: Used to play music through speakers connected via Bluetooth.

Bluetooth Pairing

Bluetooth devices need an ID to connect to another device. This process is called pairing.

Pairing is similar to an exchange of names or addresses. To pair with a device, only its name is required. Once paired, the information is saved for future use. After the two devices are paired, data can be exchanged between them.

The speed of data transfer in this case is 1 Mbps.

Healthy Practices:

- Switch off a device’s Bluetooth when it is not in use.
- Do not pair your networking device with unknown devices.

2. Near Field Communication (NFC)

- NFC is also a wireless technology used to communicate between devices.
- The range of NFC is extremely short, only around 4 cm.
- This makes NFC perfect to use when data is confidential.
  For example: bill can be paid when an NFC enabled debit card is touched to the card reader.

Every active NFC device can work in three modes:

a. NFC Card Emulation

Smart cards contain NFC technology. This enables users to make payments to restaurants, stores, and other places. Smart phones also have NFC technology and can be used as smart card.

b. NFC Reader/Writer

Devices with NFC readers can read information stored in NFC tags. NFC tags contain NFC chips. These chips are activated when an NFC enabled device is near them. This feature is commonly used in biometric readers to mark attendance of employees at workplaces or in hotel room cards.

c. NFC Peer-to-Peer

NFC peer-to-peer works when two devices are less than 4cm apart from each other. This feature is usually used by smart phone users to transfer data from one phone to another.

Healthy Practices:

- NFC technology is very secure, but no amount of security can protect someone from a stolen phone. A user must take utmost care to keep the device or card safe.

Exercise

Fill in the blanks

1. ________________ is a popular technology in smartphones to transfer files in short range.
2. The full form of NFC is ________________.

Tick mark ☑ the correct answer

3. Which of these devices uses Bluetooth?

a. CPU ☐ **b.** Motherboard ☐

c. Dongle ☐ **d.** Speakers ☐

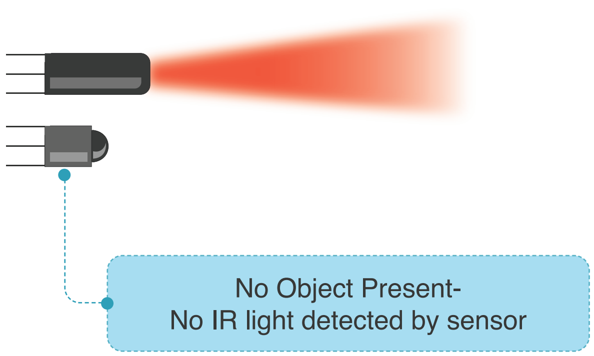
4. Which of these are uses of NFC?

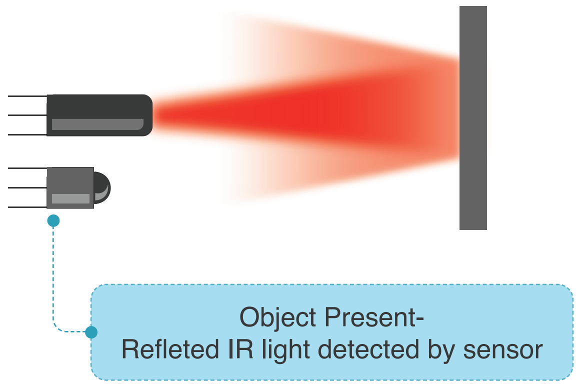
a. Making payments ☐ **b.** Buying tickets ☐

c. In identification cards ☐ **d.** All of the above ☐

3. Infrared Communication (IR)

- Infrared is a wireless mobile communication technology that works over a short range.
- Infrared doesn't require any special permission for pairing.
- Unlike other wireless communication technologies, IR cannot go past solid objects like walls. It bounces back, similar to light.





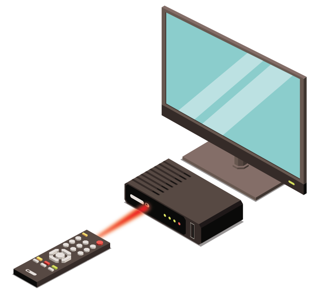
2. Wireless Communication

To use infrared communication, there should be nothing in between two devices which can obstruct its signal.

- The range of IR is a few feet, if the transmitter and receiver are in the line of sight.
- Infrared technology has limited usage for data transfer in computing devices. But it is extensively used in other applications.

These are some places where infrared technology is used:

a. TV remotes.
b. Night vision cameras that are used to take photos at night.
c. Infrared generates heat. It can be used in cooking appliances, saunas, and room heaters.



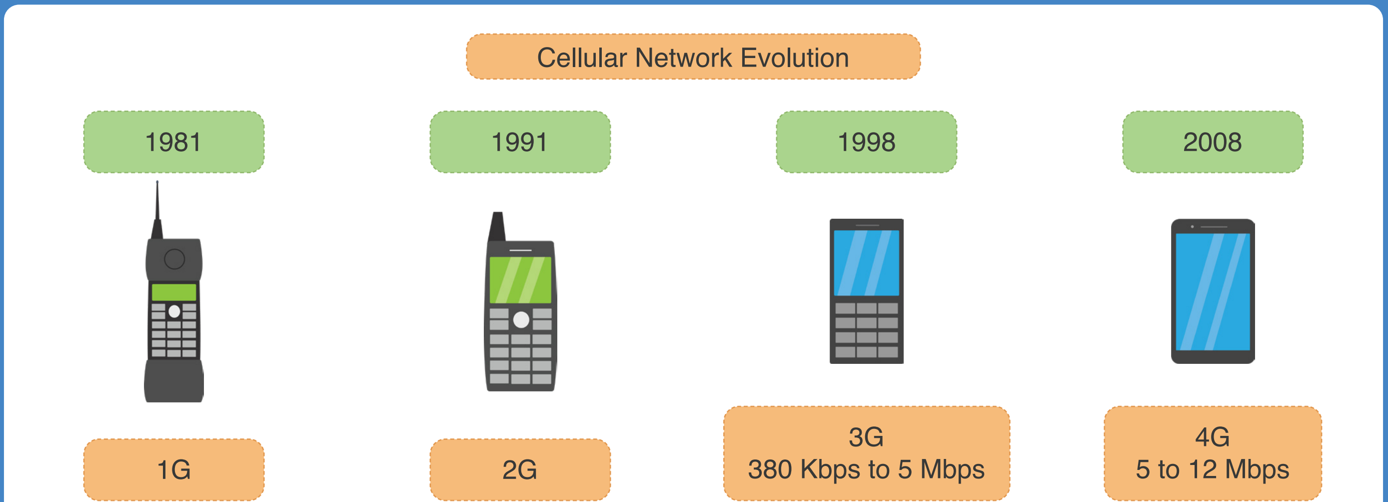
Activity

Make a list of devices in your school campus and in your home which are wirelessly connected.

Bytes

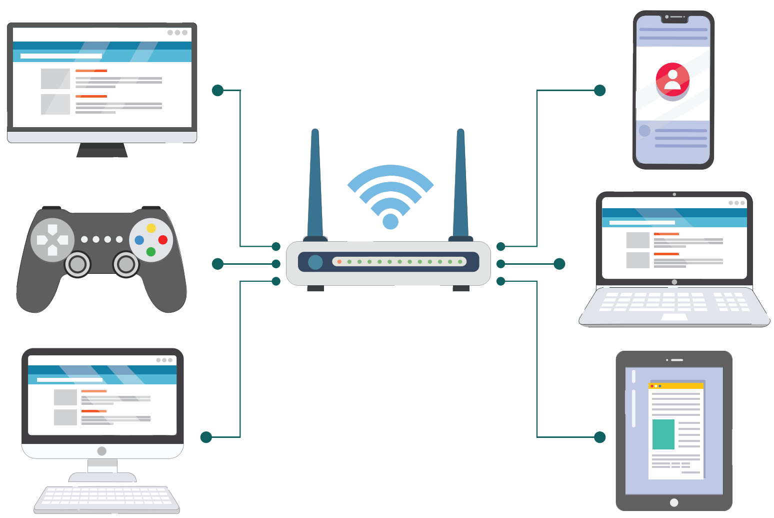
Mobile internet is classified in terms of generations.

- 1G refers to basic calls from one phone to another.
- 2G allows users to send SMS (text messages) and images via phones.
- 3G allows user to surf the internet on their mobile devices. The internet speed is between 380 Kbps & 5 Mbps.
- When 4G arrived, the internet really kicked off. This provides an excellent speed of about 5 to 12 Mbps. It allows users to stream high quality videos online.



4. Wi-Fi and Hotspot

- Wireless Fidelity or Wi-Fi allows networking devices to connect to the internet wirelessly.
- It is the most commonly used technology to access internet in today's times.
- In all the technologies we have learnt about in this lesson, devices connect to each other one-on-one.
- One Wi-Fi connection can connect to multiple devices at once.



Features of Wi-Fi

- Wi-Fi can work over large distances and connect multiple devices to the internet.
- Wi-Fi internet speeds differ from each other. The speed depends upon the network/ISP provider and internet plan subscribed to.
- Smartphones, laptops, cameras, smart speakers, and TVs are some devices that use Wi-Fi.

Benefits of Wi-Fi

- Wi-Fi can be used to access the internet from anywhere in the world.
- Wi-Fi is commonly used to stream videos, play games, and research information.
- Devices use the internet through Wi-Fi to provide services like video streams, online classrooms, gaming and virtual assistance.

Portable Wi-Fi Hotspot

a. Mobile Hotspot

A hotspot creates a Wi-Fi connection on a device that can access the internet through its mobile data. This device acts as a router and distributes data to the devices connected to it.

b. Dongle as a portable Wi-Fi hotspot

A dongle or data card is a portable Wi-Fi modem and router. A device can be wirelessly connected to it. It contains a sim card that can access the internet just like a mobile phone.

A dongle is light and compact. It can be carried anywhere.

Healthy Practices:

a. Bring your own Wi-Fi (dongle etc.) instead of using public Wi-Fi.

b. Keep your Wi-Fi connection protected with a password.

c. Turn Wi-Fi off when it's not in use.

d. Turn off file sharing options.

In this lesson, we learnt about devices that help us connect and share data without wires and cables. Wireless connections are efficient and available in most parts of the world. They truly help us bring the world closer!

Exercise

Fill in the blanks

5. A ________________ creates a Wi-Fi connection on a device that can access the internet through its mobile data.

6. Many devices can wirelessly access the internet using ________________.

7. ________________ is a portable modem and router.

Match the following

8. **a.** Wireless Internet
   b. Dongle
   c. Hotel Check-in Card
   d. Remote Control
   e. Infrared
   f. NFC
   g. Portable Hotspot
   h. Wi-Fi

2. Wireless Communication

Tech Flash

- **Range** : Range is the maximum physical distance between two networking devices to keep their connection intact.
- **Data transfer speed** : Data transfer speed is the amount of data transferred from one device to another per second.
- **Bluetooth** : Bluetooth is a wireless technology used to transfer data between short distances. It is popularly used in mobile phones and other portable devices.
- **NFC** : Just like Bluetooth, NFC is a wireless technology used to transfer data between extremely short distances. Its range is only 4 cm.
- **Infrared** : Infrared is a wireless communication technology which works over a short distance.
- **Wi-Fi** : Wireless Fidelity or Wi-Fi is a technology that allows networking devices to connect to the internet.
- **Hotspot** : Hotspot creates Wi-Fi on a device that can access internet on its own, that is with mobile data.
- **Dongle** : A dongle or data card is a portable Wi-Fi modem and router can be plugged or connect to a device wirelessly.

Future Tech

5G Wi-Fi

In coming years, internet will evolve to new generation of 5G. 5G promises a speed around 1Gbps. It is 10 times faster than present 4G.

5G capabilities now allow us to use technologies like driverless cars, advanced robots, VR experience with no lag, advanced home appliances and build smart cities.